



Deterministic and uncertainty crude oil price forecasting based on outlier detection and modified multi-objective optimization algorithm

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ABSTRACT

Crude oil price forecasting provides reference for stabilizing the energy economic, significantly improving the investment and operation decision-making. However, owing to the complexity and significant fluctuations of crude oil market, it is difficult to achieve accurate and reliable crude oil price forecasting. In this study, we develop a novel hybrid crude oil price forecasting model. Firstly, the Hampel identifier is employed to remove outliers in crude oil price series. Subsequently, complete ensemble empirical mode decomposition is used to reduce noise. Then a novel modified multi-objective water cycle algorithm is proposed to optimize the parameters of echo state network. Finally, deterministic and uncertainty prediction is conducted to verify the model performance. The results reveal that the proposed hybrid model outperforms various contract models in deterministic and interval predictions, as well as in daily and weekly forecasting. Therefore, the proposed hybrid model is a reliable tool for crude oil price forecasting and serve as a reference for decision making in energy economic market.

1. Introduction

As the crucial ingredient for industrial production activities, crude oil has profound effects on the development and stabilization of global economy and international financial markets. Crude oil price fluctuations can destabilize the global economy and restrict its growth, thereby negatively affecting the development and safety of countries globally (Wang et al., 2020b; Wang et al., 2020). Effective crude oil price prediction is essential for making and adjusting energy policies to ensure the economic stability. However, crude oil prices are determined by the result of the gaming between several dynamic and multi-dimensional factors (Wang et al., 2020b), including supply and demand (Yang et al., 2021), inventory (Yu et al., 2014), economic growth (Zhao et al., 2017), political stability (Urolagin et al., 2021), psychological anticipation, and extreme events (Yu et al., 2008b), leading to high nonlinearity and instability. Consequently, crude oil price forecasting is a challenging task. Developing reliable crude oil price forecasting methods and increasing the accuracy of prediction models to effectively reduce energy risks and stabilize the global economy have gathered increasing attention and become the most important and popular research topics.

In general, price forecasting methods can be categorized into two groups: econometric methods and artificial intelligence (AI) methods.

Traditional econometric methods, such as the vector autoregressive model (VAR) (Plakandaras et al., 2019), autoregressive integrated moving average model (ARIMA) (Xiang and Zhuang, 2013), generalized autoregressive conditional heteroskedasticity model (GARCH) (Nomikos and Andriosopoulos, 2012), and error correction model (ECM) (He et al., 2010), etc. Have been widely employed in crude oil price prediction. However, crude oil price series are highly nonlinear, nonstationary and complex, and traditional econometric methods lack the ability to capture nonlinear characteristic leading to numerous problems in parameter setting and function selection (Niu et al., 2022). Therefore, econometric methods cannot provide satisfactory forecasting performance.

Recently, the rise of AI has led to the development of numerous crude oil price forecasting methods using artificial neural network (ANN) (Huang and Wang, 2018), support vector machine (SVM) (Fan et al., 2016), deep learning (DL) (Chen et al., 2017) etc., to achieve more promising forecasting results. For example, Shambora et al. (Shambora and Rossiter, 2007) developed an ANN-based crude oil price forecasting model to serve as a reference for investors. Xie et al. (2006) conducted

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the crude oil price forecast on account of SVM. Cen et al. (Cen and Wang, 2019) applied the long short term memory DL method to predict crude oil price. These studies have demonstrated that AI methods provide better complex nonlinear time series forecasting, but there are still some challenges such as overfitting and local optimum problems that limits the performance of AI methods in forecasting (Yu et al., 2008a).

To overcome these defects of single forecasting methods, various hybrid methods that integrate the data pre-processing and optimization algorithms have been designed and applied in time series forecasting. Usually, the hybrid methods combine the superiorities of different methods, significantly improving the forecasting results (Yu and Ma, 2021). Yang et al. (2021) proposed a hybrid model based on K-means method, kernel principal component analysis (KPCA) and kernel extreme learning machine (KELM) for crude oil price forecasting, which indicates the hybrid method is better at crude oil price prediction. Bisoi et al. (2019) proposed a hybrid crude oil price prediction method using variational mode decomposition (VMD) and robust random vector functional link network (RVFLN), the empirical results approved that hybrid model significantly improves the forecasting performance. Jammazi et al. (Jammazi and Aloui, 2012) combined the harr trous wavelet decomposition (HTW) with multilayer back propagation neural network (MBPNN) to create a hybrid crude oil price forecasting model. These studies focused on developing hybrid models using data pre-processing techniques, which can drastically reduce the influence of the inherent complexity in data thereby improving the accuracy of crude oil price forecasting. Nevertheless, the shortcomings of AI methods restrict the forecasting performance of these hybrid method. Optimization algorithms enable AI methods to determine the optimal parameters, which can prevent the overfitting and local optimum problems, improving the forecasting accuracy. Zhang et al. (2021) conducted the crude oil price prediction based on a hybrid method which use ensemble empirical mode decomposition (EEMD) as well as VMD methods to decompose the original data, and then employed particle swarm optimization (PSO) to optimize the parameters of a KELM to improve the forecasting performance. Li et al. (J. Li et al., 2019) employed VMD for data preprocessing, then an SVM and a back propagation neural network (BPNN) optimized by genetic algorithm (GA) are for forecasting, the empirical results demonstrated this hybrid model could be used for crude oil price prediction. Additionally, Abdollahi et al. (Abdollahi, 2020) proposed a hybrid model that employed CEEMD for data preprocessing, and PSO for parameter optimization of an SVM that performed crude oil price forecasting and the resulting model achieved excellent forecasting results. Therefore, hybrid methods assembling the advantages of multiple methods can perform better in crude oil forecast. The methods discussed above are summarized in Table 1.

Furthermore, intelligent optimization algorithms inspired by natural phenomena have been widely employed for parameter optimization owing to their powerful convergence speed and search capabilities (Hao et al., 2021; Nie et al., 2021). Various metaheuristic algorithms such as GA (Holland, 1975), PSO (Kennedy and Eberhart, 1995), ant colony optimization (ACO) algorithm (Dorigo et al., 1996), whale optimization algorithm (WOA) (Mirjalili and Lewis, 2016), grey wolf optimizer (GWO) (Mirjalili et al., 2014), and water cycle algorithm (WCA) (Eskandar et al., 2012) have been developed by researchers. GA is a popular algorithm that is based on the genetic process of av organism. Although it has a robust global searching ability, its poor local searching ability and high calculation duration restrict its application (Ghosh et al., 2020). PSO is a widely used method that simulates the hunting behavior of birds, it is easy to use and provides fast calculation with a few parameters; however, it can be easily trapped into a local optimal solution (Ocran et al., 2021). ACO is inspired by the behavior of ants finding paths to food and has been widely employed in various optimization problems (Heidari et al., 2017), but the performance of ACO is restricted by the need to perform several experiments to determine harmonious parameter values. Disharmonious parameter values can cause the ACO to fall into local minima (Peker et al., 2013; Wang and

Han, 2021). WOA is a novel algorithm that is based on the feeding behavior of whales through a bubble net. It has attracted significant attention, but is prone to being trapped in a local optimum with the increase in the number of iterations (Jin et al., 2022; Luo and Shi, 2019). GWO simulates the hunting behavior of grey wolves, which is easy to implement and run fast, but occasionally fails to conduct sufficient exploration, resulting in a failure to find a global optimal solution (Faris et al., 2018). WCA is inspired by the water cycle process in nature, wherein water from streams and rivers flow into the sea. It provides significant advantages in different aspects, and its structure is simple and easy to implement. Moreover, the optimization process is simple, the parameters are easily adjustable, the search capability is strong, and the convergence speed and accuracy are high (Haddad et al., 2015; Osaba et al., 2018; Pahnkehkolaei et al., 2017; Sadollah et al., 2015a). Owing to these advantages, WCA has been effectively applied in various fields to solve complex non-linear problems (Yadav and Verma, 2020), constrained optimization problems (Chau et al., 2021), and other engineering and economic problems.

Owing to the superiority of hybrid methods, researchers have developed various hybrid models to forecast crude oil prices, improving the forecasting ability to some extent. Although hybrid models combine the advantages of various approaches, there are still some limitations, which hinder further improvement in the forecasting performance. For example, while preprocessing data, most methods focus on removing noise from the original data and neglect the existence of outliers, which may lead to significant forecasting errors. Although data preprocess methods such as EMD, and CEEMD can decompose outliers and enhance forecasting accuracy, the improvement is limited, therefore, researchers have performed an outlier detection (Zhang et al., 2019). Li et al. (H. Li et al., 2019) applied HI for outlier detection and correction, and then conducted CEEMD for data decomposing to perform an air quality index prediction. Guo et al. (2022) proposed a wind power prediction system using outlier correction and improved complex ensemble empirical mode decomposition with adaptive noise (ICEEMDAN) for data preprocessing. The outlier detection method assembles data preprocessing method can significantly improve the efficiency of data preprocessing. Besides, most hybrid methods focus on improving the forecasting accuracy but ignore the forecasting stability. To address these limitations, we developed a novel hybrid crude oil price forecasting model that includes an outlier test and handling method (using the Hampel identifier (HI)), a data denoising method (using CEEMD), an improved multi-objective optimization algorithm (a novel improved multi-objective water cycle algorithm (MOIWCA), which is presented herein), and an echo state network (ESN). First, the HI identifies and corrects the outliers in the original oil price data; subsequently, CEEMD extracts the useful information and removes the noise through decomposition and reconstitution; finally, the ESN, whose parameters are optimized by the MOIWCA, uses the processed data for forecasting. The main contributions of this research are:

- (1) Outliers are identified and corrected. Owing to the complexity of the crude oil market, crude oil prices are irregular and fluctuate significantly. The HI detects and corrects the abnormal points, and extracts the essential information from the original data.
- (2) An improved multi-objective water cycle algorithm (MOIWCA) is proposed herein. It uses an opposition-based learning strategy and adaptive nonlinear inertia weight to achieve effective optimization. The optimization ability before and after improvement were examined and compared against different algorithms. The results revealed that MOIWCA outperforms the multi-objective water cycle algorithm (MOWCA), as well as MOGWO and MOPSO. Furthermore, a sensitivity analysis is performed for MOIWCA to determine the best parameter settings.
- (3) A novel hybrid system containing outlier correction, data preprocessing, multi-objective optimization and forecasting technique is proposed. Based on the comprehensive and scientific

Table 1
Summary of the involved methods.

Category	Models	Results	Advantage	Disadvantage
Econometric Method	VAR (Plakandaras et al., 2019) ECM (He et al., 2010; Lanza et al., 2005) ARIMA (Xiang and Zhuang, 2013) GARCH (Nomikos and Andriosopoulos, 2012; Wei et al., 2010)	Increasing in global geopolitical risks implies accurate forecast. The ability of ECM to crude oil price is verified. ARIMA obtains good results in short-term crude oil price predict. GARCH performs differently in different crude oil market.	Econometric method can effectively identify the factors affecting crude oil price, and the prediction results are more explanatory.	Econometric method is hard to describe complex nonlinear features of crude oil price fluctuation.
AI Method	Single ANN (Huang and Wang, 2018; Movagharnejad et al., 2011) SVM (Fan et al., 2016; Khashman and Nwulu, 2011; Xie et al., 2006) DL (Cen and Wang, 2019; Chen et al., 2017) Hybrid VMD-RVFLN (Bisoi et al., 2019) HTW-MBPNN (Jammazi and Aloui, 2012) K-means-KPCA-KELM (Yang et al., 2021) VMD-EEMD-PSO-KELM (Zhang et al., 2021) VMD-GA-BPNN/SVM (J. Li et al., 2019) CEEMD-PSO-SVM (Abdollahi, 2020)	ANN can improve the forecast ability to any crude oil price. SVM is an effective candidate for forecasting crude oil prices. DL capture complex data feature and improve predict accuracy. The forecasting efficiency and accuracy are approved improved. The forecast results outperform the conventional BPNN. The approach enhances monthly crude oil price forecast accuracy. The hybrid model improves crude oil prices predict accuracy. VMD-AI methods are promising for crude oil price forecasting. The model captures nonlinearity and volatility more effectively.	Single AI Method can extract nonlinear feature of crude oil price and achieve better forecasting accuracy. Hybrid method combining the advantages of multiple methods can fully mine the data character and overcome the drawbacks of single method, significantly improve the forecasting performance.	Single AI Method is easy to trapped into overfitting and local optimum problems.

evaluation indicator, the developed hybrid model obtains great forecasting performance in crude oil price forecasting.

- (4) Strong applicability and practicality of the hybrid model is attested. Two crude oil price datasets and eleven comparison models are employed. Experiment results shows the developed hybrid model can outperforms other models no matter daily prediction and weekly prediction. The proposed model can overcome the drawbacks of short-term prediction, provide reliable prediction of longer timescales.
- (5) Conduct reliable point and interval prediction. Experiments shows that the proposed model is superior both in deterministic prediction and interval prediction. The interval forecasting is conducted based on the distribution of point forecasting error, which is facilitate to construct accurate interval so that it is able to help enhance the scientific development of crude oil market and reduce the decision-making risks.

The remainder of this paper is organized as below: Section 2 introduces the methodology of the study. Section 3 describes the general framework of the developed system. Section 4 contains the data description and empirical experiments as well as the results. Further discussions and analyses are out on in Section 5. Finally, the conclusions are presented in Section 6.

2. Methodology

In this part, the methods applied in this research is introduced, the concrete content shows as follows.

2.1. Hampel identifier (HI)

HI is a valid outlier recognizing and correcting technique, which applies the median value and the standard deviation of the median to identify the outliers (Yao et al., 2019). Given the data series: $X = \{x_1, x_2, x_3, \dots, x_n\}$, k is the numbers of data on either side of x_i in the measurement window, then the sliding window size is $2k + 1$, the local median and corresponding standard deviation can be defined as:

$$m_i = \text{median}(x_{i-k}, x_{i-k+1}, \dots, x_i, \dots, x_{i+k-1}, x_{i+k-1}) \quad (1)$$

$$\sigma_i = \kappa * \text{median}(|x_{i-k} - m_i|, \dots, |x_i - m_i|, \dots, |x_{i+k} - m_i|) \quad (2)$$

where, $\kappa \approx 1.4826$.

The outlier points can be detected by the algorithm

$$y_i = \begin{cases} x_i, & |x_i - m_i| \leq n_\sigma \sigma \\ \text{outlier}, & |x_i - m_i| > n_\sigma \sigma \end{cases} \quad (3)$$

According to the equation above, when the difference between the measured value and the local median is less than n_σ times σ , the measured value is normal and can be retained, on the contrary, when the difference exceeds n_σ times σ , the value is the outlier point and should be replaced by the local median m_i . Generally, n_σ is a given threshold value 3 (Jianzhou Wang et al., 2020a).

2.2. Complete ensemble empirical mode decomposition (CEEMD)

CEEMD modified by Yeh et al. (YEH et al., 2010) on the bases of empirical mode decomposition (EMD) (Huang et al., 1998) and ensemble empirical mode decomposition (EEMD) (Wang, 2011) is an adaptive signal preprocessing method, especially for nonlinear and non-stationary time series. The step of CEEMD can be explained as:

- Step 1 A pair of white noise $n_j(t)$ which is positive and negative is added into the initial series $x(t)$ and the generated series are and $x_j^+(t)$ respectively.

$$\begin{bmatrix} x_j^+(t) \\ x_j^-(t) \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x(t) \\ n_j(t) \end{bmatrix} \quad j = 1, 2, \dots, J \quad (4)$$

- Step 2 Execute the EMD method, and decompose the generated $x_j^+(t)$ and $x_j^-(t)$ to a finite set of Intrinsic Mode Functions (IMFs), respectively.

$$\begin{cases} x_j^+ = \sum_{i=1}^M IMF_{ji}^+ \\ x_j^- = \sum_{i=1}^M IMF_{ji}^- \end{cases} \quad (5)$$

where IMF_{ji}^+ and IMF_{ji}^- denote the i -th IMF of the j -th signals generated through adding positive and negative white noise. M represents the number of IMF.

- Step 3 Repeat the above steps N times, and by which there are corresponding IMFs obtained, the ensemble IMFs can be calculated by averaging the $2N$ IMFs, shown as:

$$IMF_j = \frac{1}{2N} \sum_{i=1}^N (IMF_{ji}^+ + IMF_{ji}^-) \quad (6)$$

where IMF_j is the j -th IMF component generated by the CEEMD method.

CEEMD method shows better performance on overcoming the mode mixing problem existing in EMD method, reducing the noise disturb and accelerating the computational speed (Xu et al., 2017). Anyway, CEEMD can effectively eliminate the influence of noise and help conduct efficient forecasting.

2.3. Modified multi-objective water cycle algorithm (MOIWCA)

Water cycle algorithm shows extensive application in many fields for its meliority in the convergence and robustness, in this section, the basic water cycle algorithm is present and the modified multi-objective water cycle algorithm is put forward. Details are as follows:

2.3.1. Water cycle algorithm (WCA)

Inspired by the process of water cycle in nature, WCA (Eskandar et al., 2012) which simulates the movement of water from rainfall to rivers and lakes then to the sea, has been conducted in optimization of complex continuous problem and discrete problem. The WCA abstracts the three considerable portions: the precipitation, confluence and evaporation, the WCA concrete algorithm calculation can be described as:

Definition 1 The initialization of precipitation

In the context of precipitation, the initial population of raindrops with size N_{pop} should be given at random. All the raindrops can be ranged and drawn into three sections on the basis of the cost function value. Define the best raindrop with minimum cost function value as Sea, consider the better raindrops with suboptimal fitness value which close to the sea as Rivers, and the last remaining raindrops are consider as Streams. The intensity of the flow which determine the raindrops flow into river or sea can be calculated as:

$$NS_n = \text{round} \left(\left| \frac{\cos t_n}{\sum_{i=1}^{N_{sr}} \cos t_i} \right| \times N_{Stream} \right), \quad n = 1, 2, \dots, N_{sr} \quad (7)$$

$$N_{sr} = N_{rivers} + 1 \quad (8)$$

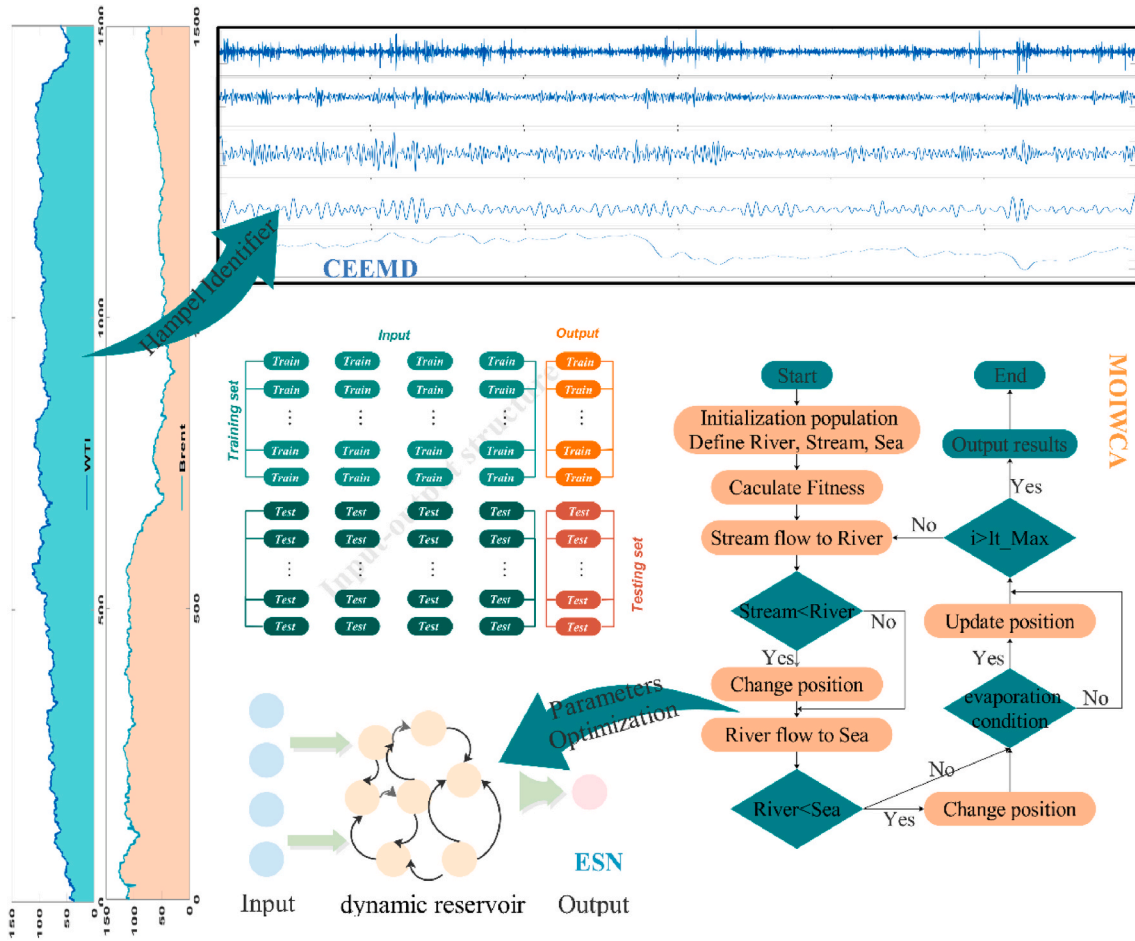


Fig. 1. The structure of the proposed hybrid model.

$$N_{Stream} = N_{pop} - N_{sr} \quad (9)$$

where NS_n represents the intensity of flow, $Cost_n$ indicates the cost function value of the n -th raindrop, N_{sr} indicates the number of river and sea, N_{Stream} is the number of streams last remaining.

Definition 2 The confluence and iteratively update

The confluence is the iteration of the optimal solution, and the position updating among Stream, River and Sea while the iteration can be calculated upon:

$$\vec{X}_{Stream}^{i+1} = \vec{X}_{Stream}^i + rand \times C \times (\vec{X}_{River}^i - \vec{X}_{Stream}^i) \quad (10)$$

$$\vec{X}_{Stream}^{i+1} = \vec{X}_{Stream}^i + rand \times C \times (\vec{X}_{Sea}^i - \vec{X}_{Stream}^i) \quad (11)$$

$$\vec{X}_{River}^{i+1} = \vec{X}_{River}^i + rand \times C \times (\vec{X}_{Sea}^i - \vec{X}_{River}^i) \quad (12)$$

where $\vec{X}_{Stream}^{i+1}$, $\vec{X}_{River}^{i+1}$ represent the new position of Stream and River, respectively; rand denotes random number selected in [0,1]; moreover, suggested value of C is 2.

The position is updated in this process. If the fitness of updated Stream is greater than River, the position of River will be replaced by Stream; Likewise, the position update between River and Sea can follow this rule, and the Sea is the optimal solution after the iteration.

Definition 3 The evaporation and precipitation

For avoiding the algorithm trapping into local optimal from rapid convergence, the evaporation and precipitation is introduced. However, only the distance between Sea and Stream or River satisfies Eq. (13), the evaporation can take place.

$$|\vec{X}_{Sea}^i - \vec{X}_{\chi}^i| < d_{max}^i, \quad \chi \in \{Stream, River\} \quad (13)$$

where the value of d_{max} is close to zero, and can be calculated as:

$$d_{max}^{i+1} = d_{max}^i - \frac{d_{max}^i}{maxiteration} \quad (14)$$

Consequently, precipitation take place and the new Stream is formed and the position is specified by Eq. (15), while the position of Stream which flow to Sea can be replaced as Eq. (16).

$$\vec{X}_{New\ Stream}^i = LB + rand \times (UB - LB) \quad (15)$$

$$\vec{X}_{New\ Stream}^i = \vec{X}_{Sea}^i + \sqrt{\mu} \times Randn \quad (16)$$

where UB and LB indicate the upper and lower limit of the variables, μ represents the constant which determines the dispersion degree of new Stream, and $Randn$ is a normal distribution matrix with the mean of 0 and variance of 1.

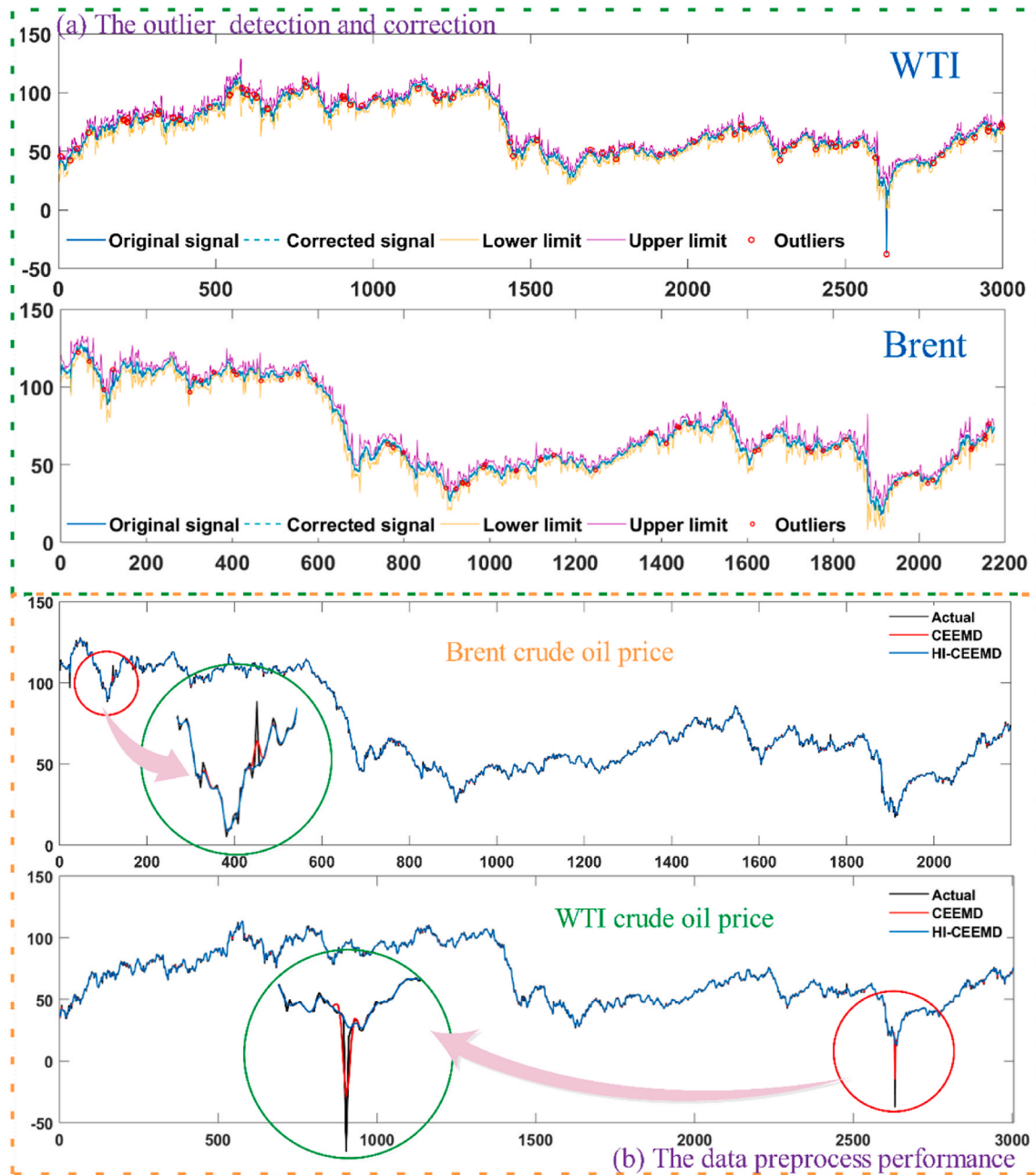
2.3.2. Improved multi-objective water cycle algorithm

Though the WCA surpasses other optimization algorithms in various field (Xu and Mei, 2018), it is still defective in some condition, and for

Table 2

The description of experimental datasets.

Dataset	Sample	Number	Mean	Max	Median	Min	Std	Kurt	Skew	ADF(p)
WTI	All Sample	3005	69.2927	113.7300	67.0400	-37.6300	21.8612	2.1195	0.0670	0.4790
	Training Set	2404	73.6529	113.7300	73.8700	26.5400	21.2867	1.7315	-0.0625	0.5318
	Testing Set	601	51.8521	76.3400	54.4700	-37.6300	14.0409	5.2827	-0.9117	0.4897
Brent	All Sample	2175	71.4888	128.3100	63.7100	17.3200	26.5379	1.9215	0.4528	0.2750
	Training Set	1740	76.4377	128.3100	68.2400	26.3900	26.6472	1.5667	0.2651	0.2207
	Testing Set	435	51.6932	76.2400	54.9800	17.3200	13.8433	2.2894	-0.4845	0.7527

**Fig. 2.** The performance of data preprocessing.

which the MOIWCA is proposed based on improvements in two aspects.

The initial solutions of the traditional WCA are generated randomly, which may bring adverse impact to the algorithm on the iteration process, rate of convergence and the performance, directly. However, the probability that the opposition solutions close to the optimal solution is

improved greatly compared with the solutions generated randomly. The opposition-based learning method (OBL) provides a way to enhance the uniformity and diversity of initial solutions, which helps enhance the solution quality and speed up the convergence speed.

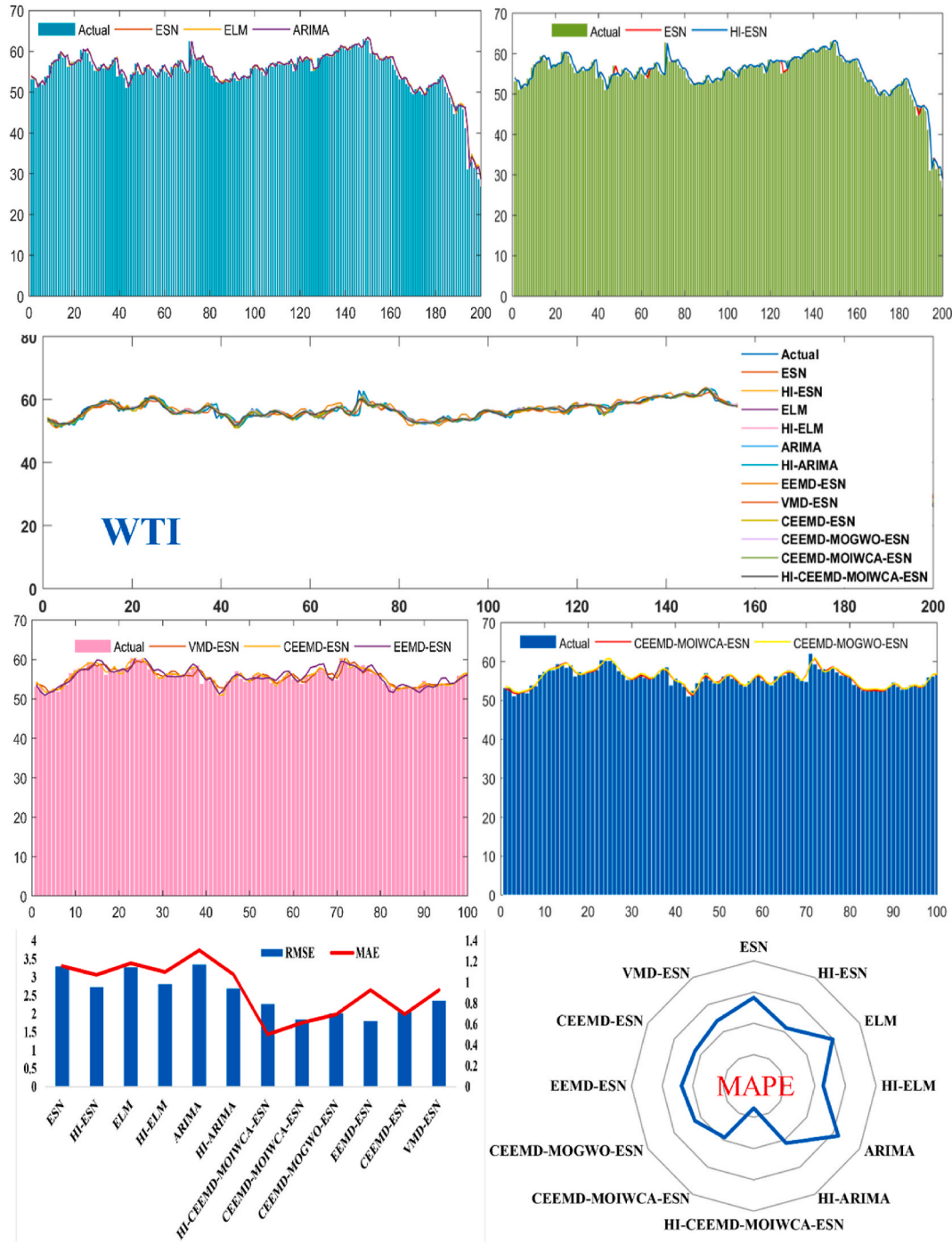


Fig. 3. The forecasting results of different models on WTI.

Definition 4 The opposition-based learning method

The opposition solutions of $X_i(x_{i1}, x_{i2}, \dots, x_{iD})$ can be defined as:

$$X'_{ij}(t) = a_j(t) + b_j(t) - x_{ij}(t) \quad (17)$$

where X is the population, N and D represent the number and dimensionality of X , x_{ij} is the population at iteration t , and $a_j(t) = \min(X_{ij}(t))$, $b_j(t) = \max(X_{ij}(t))$.

The flow of opposition-based learning is summarized as:

Step 1 Initialize the relevant parameters, containing: population size N_{pop} , individual dimension D , the upper and lower bound UB and LB .

Step 2 Generate a population randomly.

Step 3 Generate the corresponding opposition population according to Eq. (18)

$$X'_i = UB + LB - X_i \quad (18)$$

where X' is the opposition individual of X .

Table 3

Results of daily WTI crude oil price forecasting.

Models	MAE	MAPE	RMSE	U1	U2	r	IA	DA	FB
ESN	1.1565	2.8244	3.2939	0.0307	0.5850	0.9722	0.9858	0.4908	−0.0015
HI-ESN	1.0730	2.1332	2.7344	0.0255	0.9323	0.9813	0.9900	0.5395	−0.0039
ELM	1.1847	2.9815	3.2717	0.0305	0.6497	0.9727	0.9861	0.4832	−0.0006
HI-ELM	1.0998	2.2720	2.8046	0.0261	0.9381	0.9807	0.9894	0.5017	−0.0051
ARIMA	1.3095	3.1912	3.3538	0.0313	0.5740	0.9713	0.9854	0.5092	0.0007
HI-ARIMA	1.0757	2.1103	2.6924	0.0251	0.9345	0.9814	0.9904	0.4708	−0.0018
HI-CEEMD-MOIWCA-ESN	0.5025	0.7047	2.2676	0.0211	0.9186	0.9871	0.9932	0.8790	−0.0027
CEEMD-MOIWCA-ESN	0.6147	1.9050	1.8472	0.0172	0.4095	0.9914	0.9957	0.8588	0.0002
CEEMD-MOGWO-ESN	0.6950	2.2224	2.0126	0.0187	0.4360	0.9901	0.9949	0.8521	0.0014
EEMD -ESN	0.9279	2.3645	1.8014	0.0168	0.4112	0.9918	0.9959	0.6958	0.0002
CEEMD -ESN	0.6976	2.2234	2.0586	0.0192	0.3738	0.9892	0.9946	0.8017	−0.0009
VMD -ESN	0.9268	2.4031	2.3643	0.0220	0.5694	0.9858	0.9928	0.7126	−0.0003

Table 4

The daily and weekly forecasting results about Brent crude oil price.

	Method	MAE	MAPE	RMSE	U1	U2	r	IA	DA	FB
Daily	ELM	1.1149	2.6386	1.6689	0.0156	0.7249	0.9930	0.9963	0.4558	−0.0026
	ARIMA	1.0942	2.5165	1.5945	0.0149	0.7097	0.9933	0.9966	0.4988	0.0005
	ESN	1.0743	2.4814	1.5975	0.0150	0.6951	0.9934	0.9966	0.4872	−0.0009
	HI-ESN	1.0497	2.4401	1.5728	0.0147	0.6923	0.9936	0.9967	0.4825	−0.0014
	HI-CEEMD-MOIWCA-ESN	0.4408	1.0025	0.6897	0.0065	0.4540	0.9988	0.9994	0.8881	−0.0008
	CEEMD-MOIWCA-ESN	0.4923	1.1236	0.7269	0.0068	0.4471	0.9986	0.9993	0.8392	0.0002
	CEEMD-MOGWO-ESN	0.4847	1.1005	0.7194	0.0067	0.4384	0.9987	0.9993	0.8368	−0.0002
	EEMD-ESN	0.0052	1.6617	0.9790	0.0118	−0.0001	0.9975	0.7319	0.9988	−0.0001
	CEEMD-ESN	0.7349	1.6349	1.1109	0.0104	0.5435	0.9968	0.9984	0.7786	−0.0004
	VMD-ESN	1.0463	2.4206	1.4608	0.0137	0.8616	0.9945	0.9972	0.6643	−0.0002
Weekly	ELM	2.4063	5.8149	3.5021	0.0328	0.9032	0.9686	0.9828	0.5189	−0.0041
	ARIMA	2.5749	5.6820	6.8653	0.0644	0.8768	0.8851	0.9392	0.4908	0.0138
	ESN	2.3638	5.6382	3.4333	0.0322	0.8970	0.9699	0.9834	0.5259	−0.0038
	HI-ESN	2.3682	5.5963	3.4175	0.0321	0.8905	0.9697	0.9838	0.5212	−0.0016
	HI-CEEMD-MOIWCA-ESN	2.1330	5.0178	3.0802	0.0289	0.9033	0.9756	0.9869	0.5354	−0.0028
	CEEMD-MOIWCA-ESN	2.1759	5.1685	3.1516	0.0295	0.9084	0.9745	0.9863	0.5189	−0.0040
	CEEMD-MOGWO-ESN	2.1649	5.1801	3.1459	0.0295	0.9180	0.9748	0.9863	0.5283	−0.0056
	EEMD-ESN	2.2467	5.2830	3.2325	0.0303	0.8866	0.9727	0.9857	0.5142	−0.0015
	CEEMD-ESN	2.1758	5.1827	3.1407	0.0294	0.9221	0.9751	0.9862	0.5401	−0.0046
	VMD-ESN	2.2425	5.4050	3.2623	0.0306	0.8966	0.9729	0.9851	0.5236	−0.0040

Step 4 Combine the random population and the opposition population, then choose half of the individual who owes lower fitness value as the initial population.

It is known that the global and local searching ability both play vital significance on optimization algorithm, it critical to balance the two factors. An adaptive nonlinear inertia weight method is acceded to guarantee global exploration ability while avoid it trapping into local optimum. Shown as Eq. (19), ω is the weight introduced which shows positive correlation with the global searching ability and the negative correlation with the local searching capability. Consequently, an appropriate value of ω can effectively balance the global and local exploration ability, as well as promoting the performance of MOWCA. The location updating of MOIWCA can be transformed as Eq. (20). Additionally, the related concept and process of multi-objective optimization is summarized in (Sadollah et al., 2015b; Gao et al., 2022; Wu et al., 2019), and detailed information can be found there.

$$\begin{cases} \omega_{start} = 0.9, & \omega_{end} = 0.8 \\ \omega_j = \omega_{end} + (\omega_{start} - \omega_{end}) \times \exp(-3.5 \times (j/Iter_Max))^3 \\ j = 1, 2, \dots, Iter_Max \end{cases} \quad (19)$$

$$\begin{cases} \bar{X}_{Stream}^{i+1} = \bar{X}_{Stream}^i + \omega_j \times rand \times C \times (\bar{X}_{River}^i - \bar{X}_{Stream}^i) \\ \bar{X}_{Stream}^{i+1} = \bar{X}_{Stream}^i + \omega_j \times rand \times C \times (\bar{X}_{Sea}^i - \bar{X}_{Stream}^i) \\ \bar{X}_{River}^{i+1} = \bar{X}_{River}^i + \omega_j \times rand \times C \times (\bar{X}_{Sea}^i - \bar{X}_{River}^i) \end{cases} \quad (20)$$

In virtue of the opposition-based learning method and adaptive nonlinear inertia weight method, developed MOIWCA has greatly improved in performance, the quality of initial solution enhanced, the rate of convergence accelerated, the relation of global and local searching ability balanced, ultimately, the optimization to the specified parameters is more effective.

2.4. Echo state network (ESN)

As a neo-recurrent neural network proposed by Jaeger (Jaeger and Haas, 2004), ESN has been extensively adopted to dynamic pattern recognition and classification, system control and fault diagnosis, especially possesses higher prospect of application in time series forecasting (Gao et al., 2021; Han and Xu, 2015; Xu et al., 2018). Different from ecumenical neural network, the hidden layer of ESN is replaced by a structure named dynamic reservoir (DR) which is composed of plentiful sparsely connected neurons, and for which it can extract the nonlinear information lightly (Hu et al., 2021). The structure of ESN is displayed as Fig. 1.

At moment t , the state matrix of input $u(t)$, reservoir $x(t)$, and output

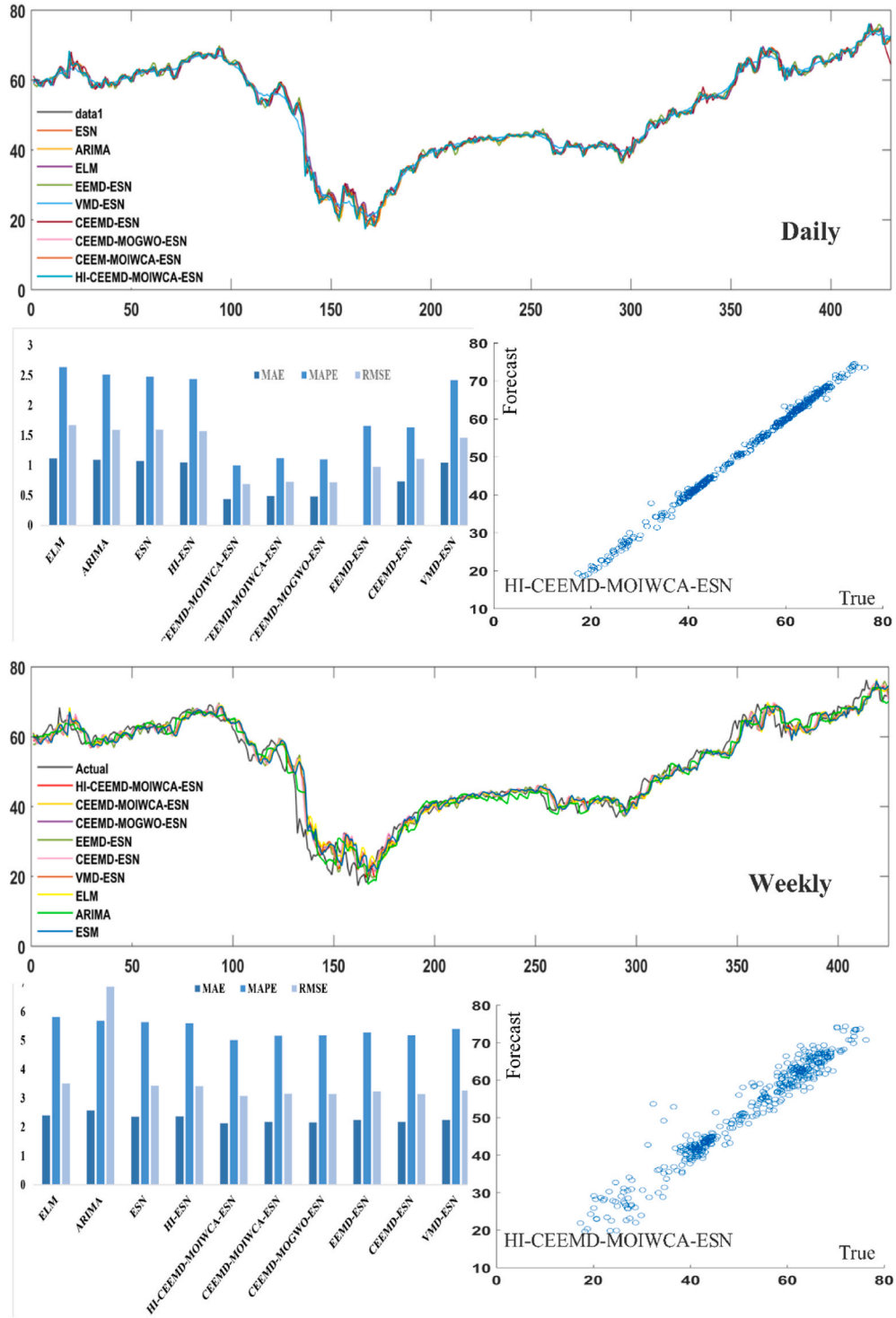


Fig. 4. The forecasting results of daily and weekly Brent crude oil price.

$y(t)$ can be expressed as:

$$u(t) = [u_1(t), u_2(t) \dots, u_K(t)]^T \quad (21)$$

$$x(t) = [x_1(t), x_2(t) \dots, x_N(t)]^T \quad (22)$$

$$y(t) = [y_1(t), y_2(t) \dots, y_L(t)]^T \quad (23)$$

where K is the unit number of input layer, N denotes that the reservoir owns N internal points, and L means the number of output unit.

The modeling process of ESN can be described as the states updating of reservoir and output Eq. (24) and Eq. (25) define the process.

$$x(t+1) = f(W_{in} \times u(t+1) + W \times x(t) + W_{back} \times y(t)) \quad (24)$$

$$y(t+1) = f_{out} \times (W_{out} \times (u(t+1), x(t+1))) \quad (25)$$

where f denotes the activation function of reservoir; W_{in} , W and W_{back} denote the connection weights matrix of input layer-reservoir, reservoir-reservoir and output layer-reservoir, respectively; f_{out} denotes the activation function of output layer; W_{out} denotes the connection weights

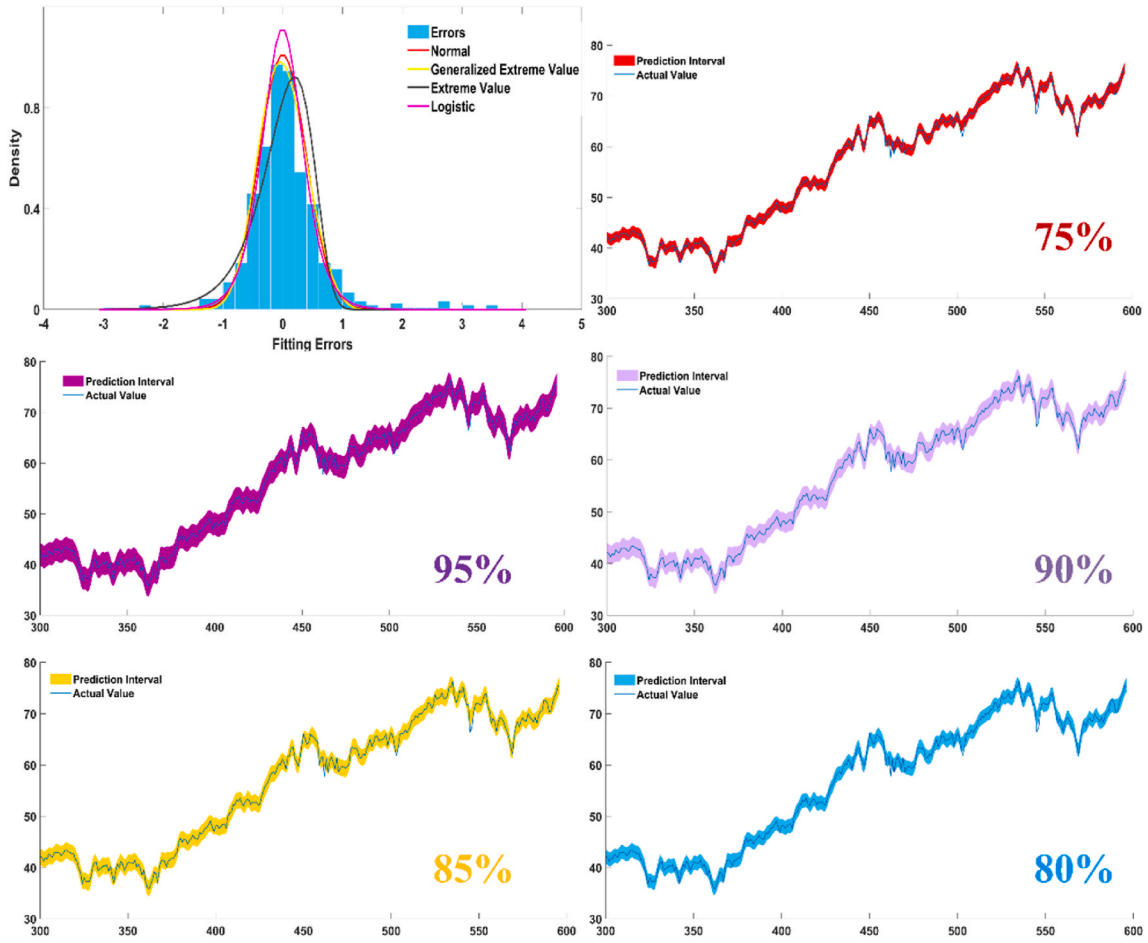


Fig. 5. The interval prediction of daily WTI crude oil price.

matrix from reservoir to output layer, and can be calculated as Eq. (26), where M is the reservoir state vector.

$$W_{out} = (M^{-1} \times y)^T \quad (26)$$

3. System development

The structure of the devised forecasting system is systematically introduced and the evaluation system for the forecasting system is described.

3.1. System design

The flowchart of the designed hybrid prediction system is exhibited in Fig. 1. The forecasting procedure can be briefly summarized as:

- (1) Perform a detailed data analysis and description, and adequately mine the primitive character from the signal to guarantee the validity of the following operation. The basic statistical characteristics including mean, max, median and min values of the original data are calculated, besides, ADF test, kurtosis and skewness are also described.
- (2) HI is applied to discern and correct the abnormal values existing in the crude oil price data, by which the weakness caused by the outlier is eliminated. Taking the initial crude oil price data to calculate the local median and standard deviation, and then identify the outliers based on '3 σ ' principle, finally, replace the outliers with the local median value.

- (3) CEEMD is conducted to reduce the interference from the noise by decomposing the corrected series and reconstitute a new series. CEEMD is performed based on decomposition and reconstruction. Add a pair white noise whose noise level is set as 0.5 to the series, and divide the series into five subseries containing four IMFs and a residual, remove the residual and add up the four IMFs as the new series. Additionally, the reconstructed price series are normalized to [-1,1] and divided into training set and testing set.
- (4) An improved multi-objective algorithm MOIWCA is proposed for optimizing the significant parameters of ESN for better forecasting performance. There are three key parameters of ESN: spectral radius, reservoir size and connectivity, which are optimized by MOIWCA. To determine the optimal parameters of the ESN, and simultaneously ensure both forecasting accuracy and stability, MOIWCA is performed with two optimization objectives:

$$\min \begin{cases} fitness_1 = \frac{1}{n} \sum_{i=1}^n (Y_{pre(i)} - Y_{act(i)})^2 \\ fitness_2 = std(Y_{pre(i)} - Y_{act(i)}) \end{cases} \quad i = 1, 2, \dots, n$$

where $Y_{pre(i)}$ and $Y_{act(i)}$ indicate the forecast results and the original value of training dataset, n is the length of crude oil price data.

As the iteration of MOIWCA, the minimum objective function values can be obtained and the optimal parameters of ESN can be output. Then update the parameters of ESN and conduct forecasting with data of the testing dataset.

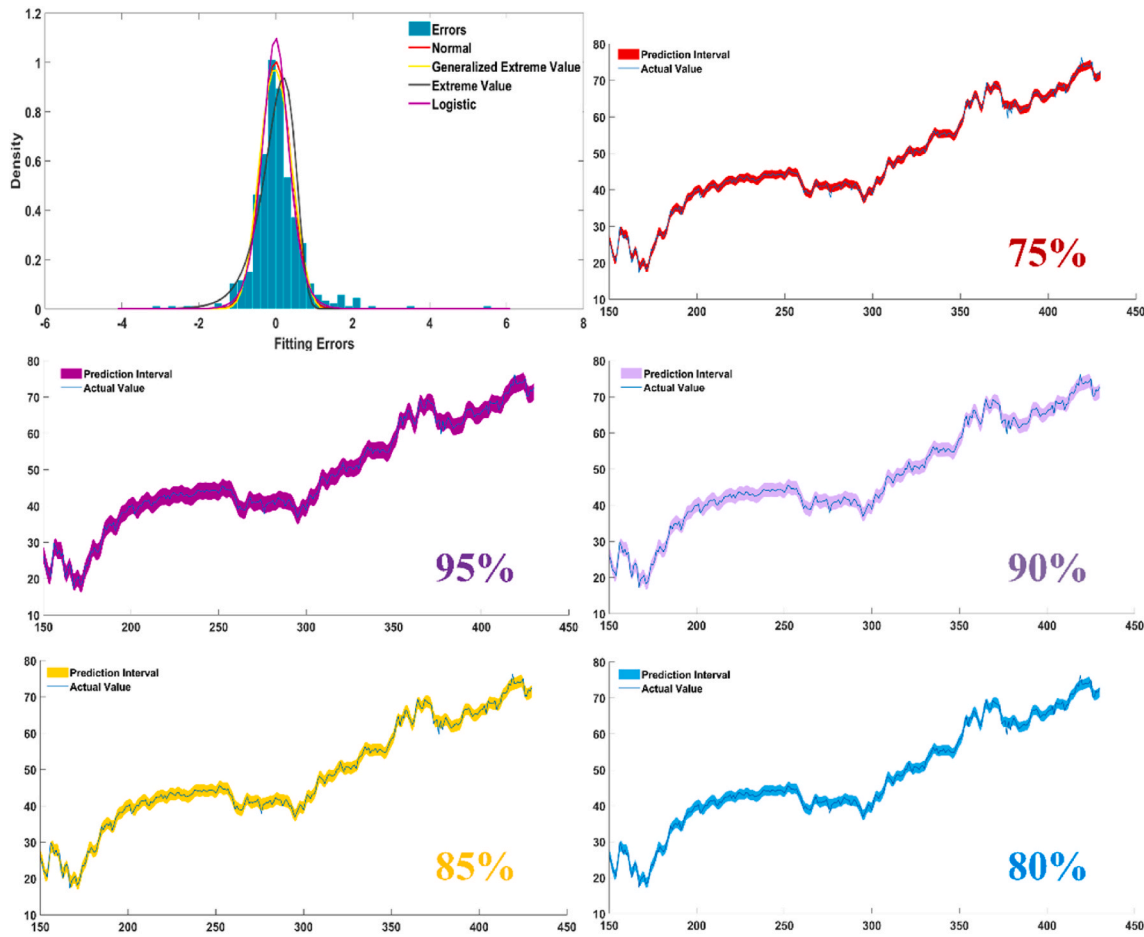


Fig. 6. The interval prediction of daily Brent crude oil price.

- (5) Various evaluation indices are applied to comprehensively assess the forecasting results, moreover, diverse forecasting techniques are considered as contrasts to confirm the superiority of the designed forecasting system.
- (6) Several discussions are performed to further demonstrate the superiority of the proposed hybrid model, containing multi-objective optimization algorithm testing and sensitive analysis, statistical test and forecasting effectiveness, moreover, multi-step forecasting and the resources policies are also added.

3.2. System evaluation

To evaluate the forecasting performance of the designed hybrid system more scientific and comprehensive, several typical evaluation indicators and interval forecasting criteria are introduced in this section.

3.2.1. Typical evaluation indicators

A total of nine typical evaluation indicators are selected to assess the forecasting results from different perspectives. The detailed computational methods of these evaluation indicators are listed in Appendix B. MAE, MAPE, RMSE, U1 and U2 are used to evaluate the forecasting accuracy, the lower the value of each indicator, the higher the forecasting accuracy. R represents the correlation between the actual and forecasting values. IA is a dimensionless indicator used to compare different methods. Furthermore, DA and FB evaluate the accuracy of the forecasting direction.

3.2.2. Evaluation indices for interval forecast

To assess the uncertainty in the forecasting, three evaluation indices

are introduced. Details are shown in Appendix B. PICP evaluates the forecasting reliability, which applies coverage rate that the built prediction interval can cover the actual observations to assess the forecasting accuracy. PINAW is applied to estimate the effectiveness of the interval prediction results, it is the valuable information amount in the prediction interval. CWC and Score are two comprehensive criteria to assess the quality of the forecasting intervals (Liu et al., 2022).

4. Empirical analysis

To verify the effectiveness of the developed hybrid model of this research, daily WTI and Brent spot crude oil price series are selected as samples, and three experiments are conducted. The specific data analysis and experiment settings as well as the forecasting results are introduced in this part.

4.1. Data analysis

Currently, the international crude oil market refers to three major crude oils (WTI, Brent, Dubai) as benchmarks. The daily WTI and daily Brent spot crude oil price data from Wind database (<https://www.wind.com.cn>) are used as experimental samples. The daily WTI data, extend from January 14, 2009 to September 29, 2021. There are 3005 observations in total, the training set (2404 observations) ranges from January 14, 2009 to June 3, 2019, accounting for 80% of the total observations, whereas the rest is the testing set (601 observations) which ranges from June 4, 2019 to September 29, 2021. The daily Brent spot data were recorded between December 28, 2011 and September 21, 2021. The training dataset comprised 1740 observations and the test

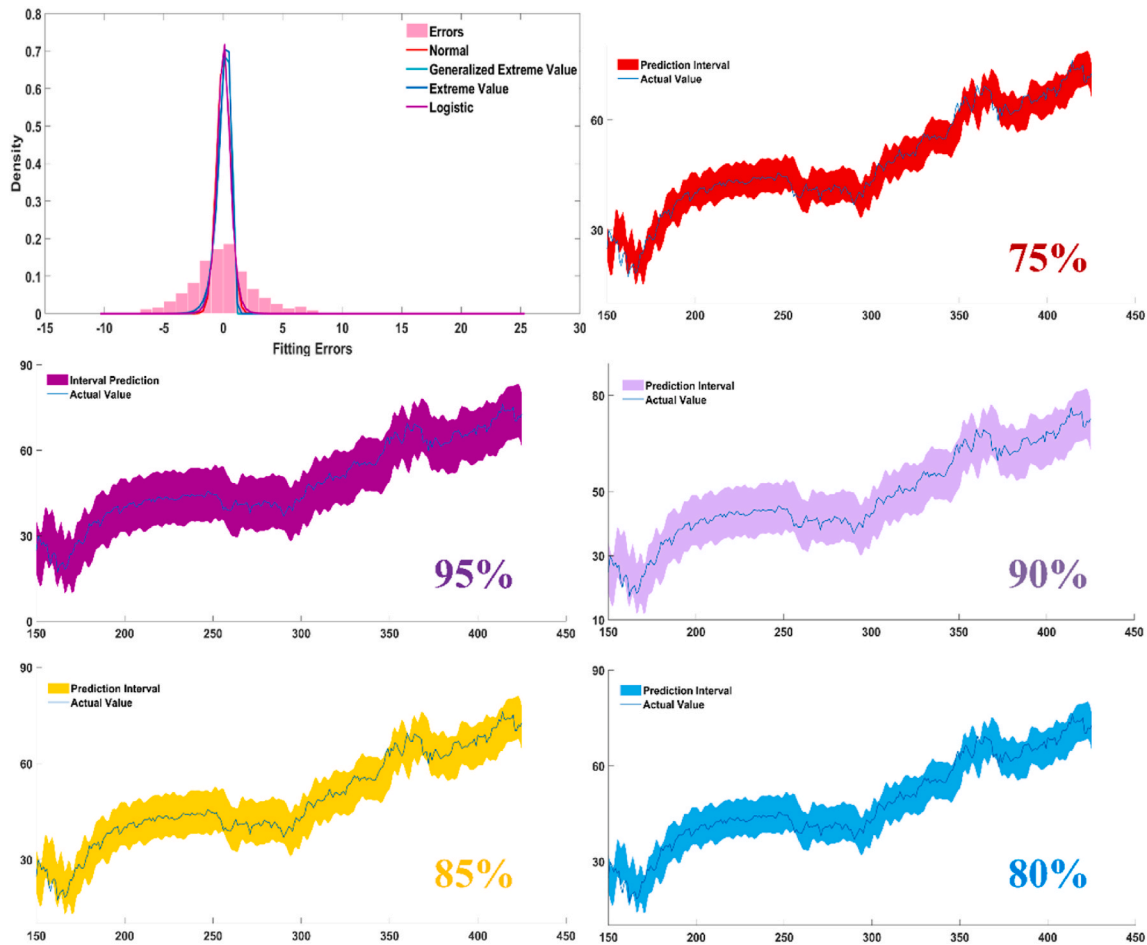


Fig. 7. The interval prediction of weekly Brent crude oil price.

Table 5

Interval prediction results for daily WTI and Brent crude oil price.

	α	95%	90%	85%	80%	75%
Daily WTI	PICP	0.9799	0.9715	0.9681	0.9631	0.9413
	PIAW	0.0391	0.0328	0.0280	0.0245	0.0193
	CWC	0.0391	0.0328	0.0280	0.0245	0.0193
	Score	0.8265	1.1658	1.4080	1.5955	1.8579
Daily Brent	PICP	0.9860	0.9744	0.9674	0.9535	0.9326
	PIAW	0.0790	0.0662	0.0565	0.0494	0.0390
	CWC	0.0790	0.0662	0.0565	0.0494	0.0390
	Score	0.5180	0.8606	1.1102	1.3110	1.5920
Weekly Brent	PICP	0.9906	0.9859	0.9718	0.9412	0.8941
	PIAW	0.3183	0.2669	0.2277	0.1991	0.1572
	CWC	0.3183	0.2669	0.2277	0.1991	0.1572
	Score	2.1084	3.4639	4.4340	5.2474	6.4973

dataset included 435 observations. The ratio of the number of data in the training and test datasets is 4:1, which was verified to be the best ratio (Yu et al., 2008b).

WTI and Brent crude oil prices are the most representative prices in the global crude oil market. From Fig. 1, the crude oil price data are nonlinear and the detailed statistical information of the original data is summarized in Table 2. The mean, max, median and min of observations are calculated, and the standard deviations of both WTI and Brent are substantial, which implies that the data fluctuate significantly. Moreover, the p values of ADF test are all greater than 0.05, indicating the data is unstable.

4.2. Experiment setting

To validate the performance of the developed hybrid forecast system, three experiments are designed. Specifically, Experiment 1 conducts a day ahead forecast based on WTI price series, Experiment 2 conducts a day ahead and a week ahead forecast based on Brent price series. The first two experiments compare the proposed system with compared models about the point forecast ability, and Experiment 3 conducts interval prediction based on the forecasting error of the first two experiments. Furthermore, the dataset is divided into training set $\{x_t^{1,\dots,t}\}$ and testing set $\{x_t^{tr+1,\dots,end}\}$, and for WTI $tr = 2404$, for Brent data $tr = 1740$, the training set is applied to train the model, the testing set is applied to validate the model. This research sets the input node as 4 and outputs a point, and the day ahead forecasting is one-step forecasting, while the week ahead forecasting is the six-step forecasting. Moreover, the length of training set is fixed and the model is not re-estimated during the forecasting process.

4.3. Performance results of data preprocessing

In this paper, HI and CEEMD method together form the data preprocessing section, the HI method conducts outlier detection and correction, while CEEMD performs noises removing. This part is used to verify the validity of the two methods.

Outliers generally indicate the points that deviate significantly from major observation. The existence of outliers may lead to incorrect parameters estimation, model errors, invalid forecasting and inferences (Liu et al., 2004). Outlier detection and correction can overcome the drawbacks and HI is an efficient outlier identifier. The application of the

Table 6
Comparisons between MOIWCA and MOPSO, MOGWO.

Function	Algorithm	Min	Average	Median	Max	std	time
zdt1	MOIWCA	0.0091	0.0118	0.0106	0.0255	0.0037	21.7436
	MOPSO	0.0114	0.0149	0.0148	0.0201	0.0021	25.8112
	MOGWO	0.0190	0.0308	0.0252	0.0721	0.0138	47.7897
zdt3	MOIWCA	0.0102	0.0184	0.0124	0.0632	0.0133	15.8311
	MOPSO	0.1275	0.1838	0.1867	0.2069	0.0156	27.4080
	MOGWO	0.0168	0.0356	0.0302	0.0833	0.0156	33.6491
dtlz2	MOIWCA	0.1745	0.1897	0.1865	0.2114	0.0094	18.4572
	MOPSO	0.1811	0.1942	0.1922	0.2198	0.0102	29.7861
	MOGWO	0.4303	0.5234	0.5136	0.5961	0.0479	75.7114
Viennet3	MOIWCA	2.6637	2.6637	2.6637	2.6638	0.0000	18.1492
	MOPSO	2.6638	2.6657	2.6639	2.7074	0.0087	27.6272
	MOGWO	2.6647	2.7782	2.7787	2.9587	0.0770	96.7933

NOTE: the bold values are the best value of the corresponding indicators.

Table 7
The comparison between MOWCA and MOIWCA.

		MOWCA					IMOWCA				
zdt1		Min	Average	Median	Max	std	Min	Average	Median	Max	std
	GD	0.0009	0.0309	0.0048	0.4194	0.0857	0.0011	0.0051	0.0042	0.0201	0.0046
	RGD	0.0092	0.0576	0.0114	0.6430	0.1464	0.0091	0.0118	0.0106	0.0255	0.0037
	S	0.0209	0.0747	0.0761	0.0949	0.0160	0.0503	0.0744	0.0741	0.0990	0.0129
	DELTA	0.3794	0.5362	0.4331	1.2921	0.2513	0.3427	0.4206	0.4005	0.5622	0.0501
zdt3		Min	Average	Median	Max	std	Min	Average	Median	Max	std
	GD	0.0054	0.0319	0.0134	0.1340	0.0384	0.0046	0.0094	0.0073	0.0407	0.0080
	RGD	0.0099	0.0590	0.0289	0.2552	0.0690	0.0102	0.0184	0.0124	0.0632	0.0133
	S	0.0771	0.2071	0.2201	0.3733	0.0889	0.1202	0.2484	0.2555	0.3349	0.0532
	DELTA	0.4642	0.8377	0.7668	1.2945	0.2579	0.4786	0.6853	0.6071	1.0813	0.1908
dtlz2		Min	Average	Median	Max	std	Min	Average	Median	Max	std
	GD	0.2674	0.2813	0.2799	0.2942	0.0069	0.2615	0.2797	0.2796	0.3016	0.0098
	RGD	0.1788	0.1911	0.1900	0.2147	0.0083	0.1745	0.1897	0.1865	0.2114	0.0094
	S	0.0637	0.1047	0.1004	0.1628	0.0243	0.0654	0.1018	0.0988	0.1481	0.0229
	DELTA	0.5866	0.7469	0.7553	0.8641	0.0557	0.6230	0.7405	0.7475	0.8162	0.0465
Viennet3		Min	Average	Median	Max	std	Min	Average	Median	Max	std
	GD	0.3379	0.3532	0.3504	0.3767	0.0102	0.3260	0.3415	0.3422	0.3518	0.0077
	RGD	2.6637	2.6638	2.6638	2.6640	0.0001	2.6637	2.6637	2.6637	2.6638	0.0000
	S	0.1118	0.1546	0.1576	0.1987	0.0228	0.1050	0.1510	0.1505	0.1934	0.0209
	DELTA	0.8575	0.8743	0.8729	0.9059	0.0109	0.8446	0.8588	0.8582	0.8781	0.0111

NOTE: the bold values are the best value of the corresponding indicators.

Table 8
Results for sensitive analysis of MOIWCA.

	Number	MAE	MAPE	RMSE	U1	U2	r	IA	DA	FB
Population Size	100	0.5660	0.8862	2.3088	0.0215	0.9157	0.9866	0.9930	0.8588	-0.0026
	150	0.5028	0.7050	2.2673	0.0211	0.9190	0.9871	0.9933	0.8773	-0.0027
	200	0.5427	0.8305	2.2878	0.0213	0.9143	0.9868	0.9932	0.8739	-0.0019
Maximum Iteration	40	0.5435	0.8264	2.2938	0.0214	0.9166	0.9868	0.9931	0.8756	-0.0024
	50	0.5028	0.7050	2.2673	0.0211	0.9190	0.9871	0.9933	0.8773	-0.0027
	60	0.5037	0.7098	2.2717	0.0212	0.9198	0.9871	0.9932	0.8739	-0.0028

HI is the first step in the proposed hybrid forecasting system. The outliers in the original WTI and Brent spot crude oil price data are identified and corrected in this step. The application of the HI is based on the ‘3 σ ’ principle: first, the local median lm_i and the standard deviation σ_i are calculated, the outliers are the points that are less than $lm_i - 3\sigma_i$, and greater than $lm_i + 3\sigma_i$. The outlier points are corrected by the corresponding lm_i values. Furthermore, the sliding window length is six, three per side, and the detailed computation is presented in Section 2.1. The identification and correction results are shown in Fig. 2 (a), this includes the original and corrected signals, the lower and upper bounds, and the outliers. The HI method can effectively distinguish the points lying outside the lower and upper bounds of the WTI and Brent spot crude oil price series. After correction, the processed data lie within the lower and

upper bounds.

CEEMD can reduce the interference from the noise by decomposing the corrected series, removing the residual sub-series and summing up the rest IMFs to reconstruct a new series. But from Fig. 2(b), it is found that CEEMD can only reduce the interference of outliers to a certain extent, and performs a slight outlier correction. The combination of HI and CEEMD can greatly correct the outliers, reduce the noise disturb, and provide effective data preprocessing.

4.4. Experiment 1: Daily WTI spot crude oil price forecasting

The daily WTI spot crude oil price data are used to compare the designed hybrid forecasting method HI-CEEMD-MOIWCA-ESN, against

Table 9
Results of DM test.

	WTI	Brent	
	Daily	Daily	Weekly
ELM	9.0054**	14.5101**	3.4640**
ARIMA	9.9003**	15.6722**	2.2875*
ESN	8.4350**	15.0240**	2.9255**
HI-ESN	17.0715**	14.7746**	2.8288**
CEEMD-MOIWCA-ESN	1.9844*	3.2022**	2.8952**
CEEMD-MOGWO-ESN	2.2521*	2.7748**	2.5175*
EEMD-ESN	5.5450**	10.9642**	2.6612**
CEEMD-ESN	2.1586*	8.5982**	2.2997*
VMD-ESN	2.1586*	8.5982**	2.2191*

NOTE: * indicates the 5% significant level; ** indicates the 1% significant level.

Table 10
Results of forecasting effectiveness.

WTI	FE1		FE2	
HI-CEEMD-MOIWCA-ESN	0.9889		0.9455	
ESN	0.9734		0.9062	
HI-ESN	0.9745		0.9147	
ELM	0.9710		0.8917	
HI-ELM	0.9734		0.9077	
ARIMA	0.9700		0.9030	
HI-ARIMA	0.9747		0.9171	
CEEMD-MOIWCA-ESN	0.9828		0.9182	
CEEMD-MOGWO-ESN	0.9802		0.9099	
CEEMD-MOPSO-ESN	0.9801		0.9083	
EEMD-ESN	0.9773		0.9191	
CEEMD-ESN	0.9809		0.9162	
VMD-ESN	0.9751		0.8964	

Brent	Daily		Weekly	
	FE1	FE2	FE1	FE2
HI-CEEMD-MOIWCA-ESN	0.9900	0.9753	0.9540	0.8956
ELM	0.9736	0.9321	0.9419	0.8598
ARIMA	0.9747	0.9390	0.9498	0.8817
ESN	0.9752	0.9380	0.9436	0.8642
HI-ESN	0.9756	0.9389	0.9440	0.8679
CEEMD-MOIWCA-ESN	0.9888	0.9731	0.9483	0.8767
CEEMD-MOGWO-ESN	0.9890	0.9734	0.9482	0.8753
EEMD-ESN	0.9834	0.9655	0.9472	0.8753
CEEMD-ESN	0.9837	0.9625	0.9482	0.8758
VMD-ESN	0.9758	0.9437	0.9459	0.8690

ESN, ELM, ARIMA, HI-ESN, HI-ELM, HI-ARIMA, CEEMD-MOGWO-ESN, CEEMD-MOIWCA-ESN, CEEMD-ESN, EEMD-ESN, and VMD-ESN. Specifically, four comparisons are performed during the experiment: Comparison I compares the individual models to select the best forecasting model; Comparison II compares the HI-forecasting and individual forecasting models to determine the efficacy of the outlier correction method; Comparison III compares hybrid models with different data preprocessing technique; and Comparison IV compares hybrid models with different optimization algorithm. As shown in Fig. 3, the

Table 11
Results of the multi-step prediction.

Forecasting Models	MAE			MAPE			RMSE		
	1-Step	2-Step	3-Step	1-Step	2-Step	3-Step	1-Step	2-Step	3-Step
ESN	1.0040	1.4837	1.8399	2.1638	3.2397	4.0212	1.4806	2.1281	2.5811
HI-ESN	0.9871	1.4491	1.8121	2.1255	3.1727	3.9697	1.4693	2.1115	2.5732
ELM	1.0064	1.4881	1.8503	2.1682	3.2432	4.0390	1.4774	2.1306	2.5925
HI-ELM	0.9940	1.4543	1.8194	2.1455	3.1802	3.9713	1.4724	2.1112	2.5759
HI-CEEMD-MOIWCA-ESN	0.4109	0.7329	1.3431	0.8928	1.5809	2.9379	0.5992	1.0518	1.8964
CEEMD-MOIWCA-ESN	0.4607	0.9546	1.5288	1.0100	2.0766	3.3344	0.6559	1.3403	2.1392
CEEMD-MOGWO-ESN	0.5239	1.0838	1.6940	1.1441	2.3635	3.7392	0.7724	1.5584	2.4158
EEMD-ESN	0.7321	1.0729	1.5551	1.5178	2.2346	3.3046	0.9354	1.4276	2.1433
CEEMD-ESN	0.5594	0.9769	1.4991	1.2002	2.1095	3.2418	0.8073	1.3946	2.1228

forecasting values of the proposed HI-CEEMD-MOIWCA-ESN are nearest to the actual values, moreover, the MAPE and MAE are significantly less than others, thus, the proposed model has better forecasting performance. As to other models, ESN is the nearest to the curve of actual values among all the single models, HI-ESN is closer than ESN, and CEEMD-ESN is nearer than EEMD-ESN and VMD-ESN, besides, the hybrid model optimized by MOIWCA is closer to the actual values than CEEMD-MOGWO-ESN. The forecasting ability of the developed hybrid model can be demonstrated by Fig. 3, and the experiment results are listed in Table 3.

Comparison I: The forecasting method is the key component of the proposed hybrid method. Accordingly, the forecasting ability of three single forecasting methods—ESN, ELM, and ARIMA—are compared. As shown in Table 3, the MAPE of ESN is 2.8244%, ELM is 2.9815%, ARIMA is 3.1912%. Furthermore, most of the indicators of ESN are better than those of ELM and ARIMA. Therefore, ESN is superior to the other two methods and has a better forecasting ability.

Comparison II: The efficacy of HI is evaluated by comparing the performances of ESN and HI-ESN, ELM and HI-ELM, ARIMA and HI-ARIMA. As shown in Table 3, the MAPE of HI-ESN is 2.1332% which is 0.6912% lower than that of ESN. Similarly, the MAPE of HI-ELM and HI-ARIMA are 2.2317% and 2.1103% respectively, both less than their corresponding single methods. Moreover, the HI-CEEMD-MOIWCA-ESN is much better than CEEMD-MOIWCA-ESN, with MAPE of 0.7047% and 1.9050%, MAE of 0.5025 and 0.6147, respectively. Therefore, HI can significantly improve the forecasting accuracy of the designed hybrid system.

Comparison III: In this comparison, the preprocessing method CEEMD is verified against VMD and EEMD. The MAPE of CEEMD-ESN is 2.2234% which is far less than the others, and the MAPE values of EEMD-ESN and VMD-ESN are 2.3645% and 2.4031%, the two MAPE are close and greater than CEEMD-ESN. Therefore, the superiority of preprocessing method CEEMD is proved, and dedicates to improving the forecasting accuracy of the proposed HI-CEEMD-MOIWCA-ESN model.

Comparison IV: Finally, to examine the efficacy of the proposed MOIWCA, the performance of CEEMD-MOIWCA-ESN and CEEMD-MOGWO-ESN are compared. As listed in Table 3, the MAE values of CEEMD-MOIWCA-ESN and CEEMD-MOGWO-ESN are 0.6147 and 0.6950, and the MAPEs are 1.9050% and 2.2224, DA are 0.8588 and 0.8521, the values of FB are 0.0002 and 0.0014, respectively. Among all the indicators, CEEMD-MOIWCA-ESN obtains better results, therefore, the developed MOIWCA can be demonstrated to improve the forecasting accuracy.

4.5. Experiment 2: Daily and weekly Brent spot crude oil price forecasting

For further verifying the superiority of the developed HI-CEEMD-MOIWCA-ESN, this part conducts daily and weekly crude oil price forecasting for Brent. Experiment results are summarized in Table 4 and Fig. 4.

From Fig. 4, it is significant that the proposed HI-CEEMD-MOIWCA-ESN achieves the best forecasting performance. For the daily

forecasting, the forecasting curve of HI-CEEMD-MOIWCA-ESN is nearest to the curve of actual values, and curves of hybrid models are closer to curve of actual values than that of individuals. Furthermore, the MAE, MAPE and RMSE of HI-CEEMD-MOIWCA-ESN are less than that of other models. Thus, the proposed HI-CEEMD-MOIWCA-ESN model can obtain more accurate forecasting results in daily crude oil price forecasting. Though the results of weekly forecasting is not good as the daily forecasting, HI-CEEMD-MOIWCA-ESN still achieves the best results among all the models. The outstanding forecasting performance of the proposed model can be demonstrated by Fig. 4.

As listed in Table 4, the proposed HI-CEEMD-MOIWCA-ESN hybrid model exhibits the best forecasting performance with an MAPE of 1.0025% among all the contract models in the daily Brent spot crude oil price forecasting. While in the single forecasting models, ESN obtains the minimum MAPE of 2.4814%, whereas ELM gets the maximal MAPE of 2.6386%, thus ESN has better forecasting ability among individual models. Considering the data preprocessing, compares the CEEMD-ESN with EEMD-ESN and VMD-ESN to demonstrate the effectiveness of CEEMD, and MAPE are 1.6349%, 1.6617% and 2.4206%, respectively, the CEEMD-ESN has the lowest MAPE, meaning the best data preprocessing performance. As to parameters optimization, the superiority of MOIWCA is proved by comparing CEEMD-MOIWCA-ESN with CEEMD-MOGWO-ESN. Moreover, the performance of outlier correction is proved by comparing ESN with HI-ESN, or comparing CEEMD-MOIWCA-ESN with HI-CEEMD-MOIWCA-ESN. Through the comparisons, the methods used in the proposed hybrid model exhibited the best performance compared to the other models.

The forecasting performance of the weekly Brent spot crude oil price forecasting is inferior to that of daily forecasting, but the results are consistent: the developed HI-CEEMD-MOIWCA-ESN hybrid model outperforms all the contrast models, ESN performs best in individual models, hybrid strategy helps improve the forecasting results, and the algorithms employed in the proposed model for data preprocessing, parameter optimization, and outlier correction outperformed their counterparts.

Therefore, the proposed hybrid model HI-CEEMD-MOIWCA-ESN combines the advantages of multiple methods and exhibits an excellent ability to forecast daily and weekly Brent spot crude oil prices. The algorithms used in the hybrid model and the hybrid strategy can significantly improve the forecasting performance and achieve accurate and reliable forecasting results.

4.6. Experiment 3: Uncertainty analysis

Uncertainty prediction of crude oil prices is highly relevant for the crude oil industry and can provide information that point forecasting cannot. The uncertainty and potential risks in crude oil market can be quantified effectively through forecasting interval (Wang et al., 2022). This part conducts the interval forecasting based on the point prediction to obtain the interval consist of lower and upper bounds, and the results can provide more useful information (Wang et al., 2020d). From Fig. 5, Fig. 6, Fig. 7 the point forecasting error behave according to logistic distribution, and the intervals are established based on fitting errors with logistic in this research, the interval forecasting results are given as Table 5. The results differed under different confidence levels α , taking WTI as example, the PICP is 0.9799 and 0.9715 at confidence level of 95% and 90%, respectively, the corresponding PIAW is 0.0391 and 0.0328, CWC is 0.0391 and 0.0328, Score is 0.8265 and 1.1658. The results of daily and weekly Brent interval forecast are also shown there. All the PICPs are near to 1 at confidence level of 95%, means that the forecasting results are wonderful. The figures show the interval prediction results intuitively.

5. Discussion

In this paper, based on the improved multi-objective optimization

algorithm MOIWCA, two discussions are conducted.

5.1. Test of the modified optimization algorithm

To validate the optimize ability of the improved MOIWCA algorithm, four performance evaluation indexes are introduced. Generational distance (GD) (Nie et al., 2020; Van Veldhuizen and Lamont, 1998) indicates the distance between the obtained Pareto front and the true Pareto front, which represents the degree of deviation from the true boundary and the greater value means the further departure from the optimal boundary, the convergence is worse. Spacing (SP) (Bansal, 2019; Schott, 2005) estimates the standard deviation of the minimum distance from each solution to the other solution, SP is an indicator on the distributivity of solutions and the smaller the SP, the more homogeneous distribution of the solution. Diversity metric (Deb) (Deb et al., 2002) represents the spread of the solution and the smaller value, the better spread. Inverted generational distance (IGD) (Mirjalili et al., 2016; Reutin, 2011) measures the combination property of the algorithm, and the smaller value of IGD, the better performance of the algorithm. Moreover, four classical test functions are selected for simulation, the equations of functions are listed in Appendix C.

For further verifying the superiority of the improved MOIWCA, the original MOWCA and other optimization algorithms such as MOGWO and MOPSO are selected as contrast. The experiments are conducted 25 times for each comparison and the parameter settings are: 100 iterations, a population size of 50 and an achieve size of 50. The test results are shown in Table 6 and Table 7 the conclusion is summarized as follows.

Comparison I: The first comparison is conducted between the original MOWCA and the modified MOIWCA, conclusions are drawn by four test functions and four evaluation indexes. Obviously, there are more bold values falling into the part of MOIWCA, and just several bold values show in MOWCA part, which means the MOIWCA obtains more best values among the testing process and MOIWCA outperforms MOWCA. Specifically, MOIWCA gains the best average results and the best standard deviations for each test function, which indicates the MOIWCA owns better optimization ability and stability.

Comparison II: In the comparison, two widely used multi-objective optimization algorithm MOPSO and MOGWO are selected as contrast, and the results indicate that MOIWCA is superior to the contrast algorithm at the combination property. Specific to the concrete results, MOIWCA obtains the best minimum and average values which indicates that MOIWCA shows the ability to get better optimization results. Furthermore, the operation time of MOIWCA is far less than the other two algorithm so IMOWCA gains advantages in computation speed.

Remark: Modifying on MOWCA leads to great improvement of MOIWCA on optimization ability, achieving better optimization effect not only compared with MOWCA but also other optimization algorithm. Therefore, MOIWCA has excellent optimization ability and should be widely applied for multi-objective optimization problem.

5.2. Sensitivity analysis

Parameter setting is a crucial part of the performance for optimization algorithm, and for MOIWCA, population size and maximum iteration are the key parameters. The suitable parameters can promote the optimization ability of the algorithm, and if the parameters are not appropriate, the performance of the algorithm will reduce. To achieve the best crude oil forecasting performance, this part conduces sensitivity analysis of MOIWCA to explore the best parameters setting. The results are presented at Table 8.

From the experiment results, when the iteration is fixed, the forecasting results changes with the changing of population size, and when the population size enlarges, the forecasting performance improves firstly, until reaches a high level it then goes down. When the population size is fixed, the forecasting performance changing with iteration shows

the same law as population size. In this paper, the best population size is 150 and the best maximum iteration is 50, which is used in the experiments of this research.

5.3. Hypothesis testing

Diebold Mariano (DM) test (Diebold and Mariano, 1995) is the significant hypothesis testing method which verifies the significant difference between the proposed method and the contract methods to further prove the superiority of the developed hybrid method. The detailed introduction of DM test can follow the literatures (Jiang et al., 2021; Yang et al., 2022). If the DM test results beyond $[-Z_{-\alpha/2}, Z_{-\alpha/2}]$, the null hypothesis is rejected at significant level α , which means the proposed model is significantly different from the contract model. As the results shown in Table 9, all the results of DM test are larger than 1.96, where $Z_{0.05/2} = 1.96$, and most results are larger than 2.58 where $Z_{0.01/2} = 2.58$, meaning that the developed hybrid model HI-CEEMD-MOIWCA-ESN is significantly superior to the other contrasted models.

5.4. Forecasting effectiveness of the developed forecast system

To further evaluate the robustness of the developed hybrid prediction system, forecasting effectiveness (FE) is introduced, and the detailed theory can be found in (Yang et al., 2019). Additionally, the larger results indicate a better forecasting effectiveness (Lv et al., 2022b), and the one-order (FE1) and two-order (FE2) results are listed in Table 10. From the results, the proposed hybrid system obtains the best FE1 and FE2 for the daily WTI price forecast and both the daily and weekly Brent price forecast. Since the higher FE value corresponds to higher forecasting robustness, the proposed hybrid predict system achieves the best robustness among all the compared models.

5.5. Multistep prediction based on the developed model

To further validate the forecasting performance of the developed crude oil price forecasting model, a multi-step prediction is performed. This experiment is conducted based on data of Dubai crude oil price, ranging from January 14, 2009 to September 21, 2021. Besides, the parameters of models are the same as models for WTI and Brent crude oil price prediction. The results are listed in Table 11, it can be found that the developed hybrid model achieves the best forecasting results in 1-Step, 2-Step and 3-Step prediction. Compared with the other models, the superiority of the developed HI-CEEMD-MOIWCA-ESN in multi-step crude oil price prediction is demonstrated.

5.6. Policy implications

Resources policy are essential to the development of the energy consumption, economy, technology, and stability of a country. Crude oil prices are affected by policy, and in turn also influence policy owing to their volatility. A scientific, reasonable, and effective resource policy can improve the stability of crude oil prices. Considering the current complexities internationally and the volatility of crude oil prices, we offer the following policy suggestions: (1) Accelerate reform of energy structure, develop renewable energy, reduce the dependence on crude oil. With the utilization of renewable energy, the demand for crude oil will be mitigated and the oil price fluctuations will bring less impact and challenge to the social economy and stability. (2) Strengthen the development of science and technological innovation, increase the crude oil efficiency. The increasing of utilization can efficiently reduce the cost

and relieve the burden of crude oil supply and demand. (3) Enhance the monitoring of international oil price, political and economic situation, and respond to the external uncertainties timely. The crude oil price is affected by various factors and even a small variation may generate a huge impact on oil price. (4) Reinforce the supervision and management of the crude oil financial market, improve the trading system and environment, reduce the financial risks and maintain the financial market stability.

6. Conclusion

Fluctuation in crude oil price brings great challenge to the development and stability of global economic, accurate and stable future price forecasting is essential for the stable operation of crude oil market. This paper developed a hybrid framework, in which Hampel identifier is employed to identify and correct outliers, the complete ensemble empirical mode decomposition performs noise removing through decomposition and reconstruction of data. To improve the forecasting accuracy and stability, we propose a modified multi-objective water cycle algorithm, which is demonstrated to be splendid compared with other optimization algorithms. The proposed HI-CEEMD-MOIWCA-ESN is employed for daily WTI crude oil price as well as daily and weekly Brent crude oil price prediction, with MAPE values of 0.7047%, 1.0025% and 5.0178%, respectively. The results of various experiments indicate that the developed hybrid framework significantly outperforms the contrast models. Subsequently, interval prediction is conducted, and the interval forecasting results are all excellent for daily WTI, daily and weekly Brent prediction. The corresponding values of PICP at 95% confidence level are 0.9799, 0.9860 and 0.9906, PIAW are 0.0391, 0.0790, 0.3183, all the PICP values are approximate to 1, and PIAW values are smaller, thereby the proposed hybrid model can engender high quality prediction interval for crude oil price.

The hybrid model developed in this paper achieves excellent deterministic and uncertainty prediction performance for daily and weekly crude oil price forecasting. It can be an effective support for providing invest reference and mitigating risks in crude oil market. Moreover, it also can be applied in other fields, such as stock market, energy market, etc. Of course, there are still some limitations existing in this research, influence factors, for example supply and demand, political factors, exchange rate, etc are not considered, additionally, this research only conducted daily and weekly forecasting, and long-term crude oil price forecasting is ignored, these lead to more research in the future.

Credit author statement

Chunying Wu: Conceptualization, Methodology, Software, Formal analysis, Supervision, Writing-original draft. **Jianzhou Wang:** Data curation, Investigation, Methodology, Software, Visualization. **Yan Hao:** Conceptualization, Methodology, Software, Writing-review & editing.

Declaration of competing interest

The authors declare that there is no conflict of interest with regard to the publication of this paper.

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Appendix A

List of abbreviations

AI	artificial intelligence
VAR	vector autoregressive model
ARIMA	autoregressive integrated moving average model
GARCH	generalized autoregressive conditional heteroskedasticity model
ECM	error correction model
ANN	artificial neural network
SVM	support vector machine
DL	deep learning
EMD	empirical mode decomposition
WPD	wavelet packet de-noising
VMD	variational mode decomposition
RVFLN	robust random vector functional link network
HTW	harr trous wavelet decomposition
MPNN	multilayer back propagation neural network
EEMD	ensemble empirical mode decomposition
KELM	kernel extreme learning machine
KPCA	kernel principal component analysis
PSO	particle swarm optimization
GA	genetic algorithm
BPNN	back propagation neural network
HI	Hampel identifier
CEEMD	complete ensemble empirical mode decomposition
MOIWCA	improved multi-objective water cycle algorithm
ESN	echo state network
ADF	Augmented Dickey-Fuller test

Appendix B

Metric	Definition	Equation
MAE	Mean absolute error	$MAE = \frac{1}{n} \sum_{i=1}^n a_i - f_i $
MAPE	Mean absolute percentage error	$MAPE = \frac{1}{n} \times \sum_{i=1}^n (a_i - f_i) / a_i \times 100\%$
RMSE	Root mean-square error	$RMSE = \sqrt{\frac{1}{n} \times \sum_{i=1}^n (a_i - f_i)^2}$
U1	Theil U statistical 1 of forecast result	$U1 = \sqrt{\frac{1}{n} \times \sum_{i=1}^n (a_i - f_i)^2} / \sqrt{\frac{1}{n} \times \sum_{i=1}^n f_i^2} + \sqrt{\frac{1}{n} \times \sum_{i=1}^n a_i^2}$
U2	Theil U statistical 2 of forecast result	$U2 = \sqrt{\frac{1}{n} \times \sum_{i=1}^n ((f_{i+1} - a_{i+1}) / f_i)^2} / \sqrt{\frac{1}{n} \times \sum_{i=1}^n ((f_{i+1} - a_i) / f_i)^2}$
r	Pearson's correlation coefficient	$r = (\sum a_i f_i - \sum a_i \sum f_i / n) / \sqrt{(\sum f_i^2 - (\sum f_i)^2 / n)(\sum a_i^2 - (\sum a_i)^2 / n)}$
IA	Index agreement	$IA = 1 - \sum_{i=1}^n (f_i - a_i)^2 / \sum_{i=1}^n (f_i - \bar{a} + a_i + \bar{a})^2$
FB	Fractional bias	$FB = 2(\bar{a} - \bar{f}) / (\bar{a} + \bar{f})$
DA	Direction accuracy	$DA = \frac{1}{n} \times \sum_{i=1}^n m_i \times 100\%, m_i = \begin{cases} 1, & (f_{i+1} - a_i)(a_{i+1} - a_i) > 0 \\ 0, & otherwise \end{cases}$
PICP	Prediction interval coverage probability	$PICP^a = \frac{1}{n} \sum_{i=1}^n X_i, X_i = \begin{cases} 1, & f_i \in [L_i^a, U_i^a] \\ 0, & f_i \notin [L_i^a, U_i^a] \end{cases}$
PIAW	Prediction interval normalized average width	$PINAW^a = \frac{1}{nR} \sum_{i=1}^n (U_i^a - L_i^a)$
CWC	Coverage width criterion	$\begin{cases} CWC^a = PINAW^a + \varphi(PICP^a) e^{-\eta(PICP - \mu)} \\ \varphi(PICP^a) = \begin{cases} 0, & PICP \geq \mu \\ 1, & PICP < \mu \end{cases} \end{cases}$
Score	Metric Score	$Score^a = \frac{1}{n} \sum_{i=1}^n S_i^a, S_i^a = \begin{cases} -2\alpha(U_i^a - L_i^a) - 4(L_i^a - f_i), & f_i < L_i^a \\ -2\alpha(U_i^a - L_i^a), & f_i \in [L_i^a, U_i^a] \\ -2\alpha(U_i^a - L_i^a) - 4(f_i - U_i^a), & f_i > U_i^a \end{cases}$

Appendix C

ZDT1	ZDT3
$\text{Minimize} \begin{cases} f_1(x) = x_1 \\ f_2(x) = g(x) \times h(f_1(x), g(x)) \end{cases}$ $\text{s.t.} \begin{cases} g(x) = 1 + \frac{9}{N-1} \sum_{i=2}^N x_i \\ h(f_1(x), g(x)) = 1 - \sqrt{f_1(x)/g(x)} \end{cases}$	$\text{Minimize} \begin{cases} f_1(x) = x_1 \\ f_2(x) = g(x) \times h(f_1(x), g(x)) \end{cases}$ $\text{s.t.} \begin{cases} g(x) = 1 + \frac{9}{29} \sum_{i=2}^{30} x_i \\ h(f_1(x), g(x)) = 1 - \sqrt{\frac{f_1(x)}{g(x)}} - \left(\frac{f_1(x)}{g(x)}\right) \times \sin(10\pi f_1(x)) \end{cases}$
DTLZ2	Viennet3
$\text{Minimize} \begin{cases} f_1 = (1 + g(x)) \times \cos\left(x \times \frac{\pi}{2}\right) \times \cos\left(y \times \frac{\pi}{2}\right) \\ f_2 = (1 + g(x)) \times \cos\left(x \times \frac{\pi}{2}\right) \times \sin\left(y \times \frac{\pi}{2}\right) \\ f_3 = (1 + g(x)) \times \sin\left(x \times \frac{\pi}{2}\right) \end{cases}$ $\text{s.t. } g(x) = (x - 0.5)^2 + (y - 0.5)^2$	$\text{Minimize} \begin{cases} f_1(x, y) = 0.5 \times (x^2 + y^2) + \sin(x^2 + y^2) \\ f_2(x, y) = \frac{(3x - 2y + 4)^2}{8} + \frac{(x - y + 1)^2}{27} + 15 \\ f_3(x, y) = \frac{1}{(x^2 + y^2 + 1)} - 1.1 \times \exp(-x^2 - y^2) \end{cases}$

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